

The Phase Noise and Clock Synchronous Carrier Frequency Offset based RF Fingerprinting for the Fake Base Station Detection

Arslan Ali
Institute for Electronics
Engineering
Friedrich Alexander Universität
Erlangen Nürnberg
Erlangen, Germany
arslan.ali@fau.de

Georg Fischer
Institute for Electronics
Engineering
Friedrich Alexander Universität
Erlangen Nürnberg
Erlangen, Germany
georg.fischer@fau.de

Abstract—This paper presents very promising radio frequency (RF) fingerprinting techniques for the terminal-based detection of fake base station (FBS) in a wireless cellular network. The proposed schemes are based on identifying the analog hardware impairments of the transmitter. In this regard, phase noise analysis and its measured signatures reveal that low to medium end software defined radio (SDR) acting as FBS can certainly be detected if the detecting terminal's own phase noise is at least 10-dB better than the transmitter. However, for medium to high end FBS, a computationally efficient network synchronized carrier frequency offset (CFO) approach has been proposed and the measurement results confirm that the regular base stations (RBS) that are clock synchronized show identical CFO values whereas, the FBS show large and random offset values. Furthermore, the CFO stability (Frequency offset vs. time) signatures illustrate that an FBS running by its own lazy clock show a large instability in the frequency offset values and even in case of a precision clock, fluctuations vanish though the stabilized offset is still large enough for a user terminal to distinguish the FBS from the RBS.

Keywords—phase noise, carrier frequency offset, RF Fingerprinting, fake base station, software defined radio, carrier frequency stability, hardware impairments, physical layer security

I. INTRODUCTION

With the advent and evolution of software defined radio (SDR) platforms and various mobile communication standards, wireless security has become quite challenging since intruders can quickly deploy **fake base station (FBS)** by configuring an SDR with at least one of the current wireless mobile communication standards and mimicking it as a **legitimate regular base station (RBS)** [1]. As a result, adversaries may attempt to gain an unauthorized access to the user terminals and interrupt the information flow [2]. In contrary to the conventional higher layer security protocols e.g. cryptography, physical (PHY) layer security has gained a particular attention to tackle such FBS activities including RF spoofing [3], protocol-aware jamming for LTE/LTE-A [4], denial of service (DoS), eavesdropping and masquerade attacks [5-6].

In order to detect an FBS, a compressed sensing based PHY layer approach has been reported in [7] that defines the Signal-to-Interference-plus-Noise Ratio (SINR) as a key metric to detect the intrusion. However, the detection algorithm is

computationally very complex and requires a centralized monitor node (MN) in a network that calculates the SINR metric iteratively. Additionally, this is targeted particularly for Wi-Fi attacker scenarios and it is very difficult to realize this sparse detection concept to cellular networks without making large adaptations by the Mobile Network Operators (MNOs). In a similar context, a theoretical location-based approach has been proposed in [8] to identify the fake cells. This method is based on monitoring the average received synchronization signal strength (ARSSS) by assuming that FBS transmits at sufficiently higher power levels than the RBS. Nevertheless, this assumption is not necessarily true and even with low to moderate transmission power, a user terminal can still get camped to the FBS during the cell selection stage. Besides, the detection scheme relies on GPS signal and in case of GPS disconnection, adversary can easily get through. To mitigate these problems, a more realistic approach called FBSleuth [9] has been developed to detect the FBS using RF Fingerprinting. Despite the promising results, it suffers from serious limitations as it is targeted for high mobility GSM use cases only and considers the training sequences by completely ignoring the real payload data for detection algorithm. Additionally, it considers a very low performing FBS i.e. rms EVM of up to 28% and in case of medium to high end FBS, where the rms EVM should fall dramatically below 15%, FBSleuth would then be unable to tackle the adversary. The transmitter power of FBS has been considered to be 50W (+47dBm) whereas in real scenarios, SDR platforms are limited to around +15-20dBm of output power [10]. In order to increase the output power of an SDR, an external Power Amplifier (PA) must be integrated that would also boost the non-linearities along with the signal amplification. Furthermore, it also assumes the I-Q based FBS transmitter architecture and the metrics like I-Q offset and I-Q imbalance become invalid for other types of FBS architectures. Likewise, the proposed machine learning classification algorithm is computationally complex to simply detect a low performing FBS only.

In order to tackle the aforementioned issues, this paper presents an FBS detection solution which is applicable to all mobile communication standards i.e. 2G/ 3G/ 4G/ 5G and particularly projected to tackle all kinds of adversary models i.e. low, medium and high performing FBS. The detection schemes are based on a very meaningful and practical RF Fingerprinting

techniques that are both computationally efficient and fast enough to detect the attacker instantly regardless of the deployed transmitter architecture type for the FBS. The first proposed detection approach involves computation of phase noise signatures and comparing them with the standard signatures of the RBS. With the phase noise analysis, probability of detecting a low to medium performing FBS is high, however, it's difficult to detect an intrusion if high quality oscillator has been deployed in the FBS architecture. To resolve this problem, a novel approach employing a simple computation of carrier frequency offset (CFO) and frequency stability in a clock synchronous regular mobile network has been proposed. Since the legitimate BS and terminals are all synchronized to the same clock domain, CFO values for different RBS determined at a user terminal have proven to be identical and in a certain range of tolerance. With the help of a measurement setup containing low to high range SDR platforms, it has also been confirmed that an FBS running by its own lousy clock show large and instable CFO values. When fed by a precise clock, an FBS shows stability in the CFO trend, however the stable CFO is still found to be considerably large enough enabling the terminal to differentiate an FBS from the regular one.

The rest of the paper is organized as follows. Section-II presents the proposed BS RF fingerprinting schemes, section-III describes the measurement setup to demonstrate the detection schemes followed by the results and analysis in section-IV. Finally, section-V concludes the paper by drawing an outlook for the future work.

II. BASE STATION RF FINGERPRINTING SCHEMES

A. Phase Noise Signatures

The frequency synthesizer of a base station containing the oscillator and phase locked loop (PLL) analog hardware is characterized by the single side band (SSB) phase noise $\mathcal{E}(f)$ which is given by [10 - 11]:

$$\mathcal{E}(f) = 10 \log \frac{P_{n1H}}{P_c} \quad (1)$$

where P_{n1H} is noise power in a 1Hz bandwidth and P_c is the total carrier power. $\mathcal{E}(f)$ is measured in dBc/Hz.

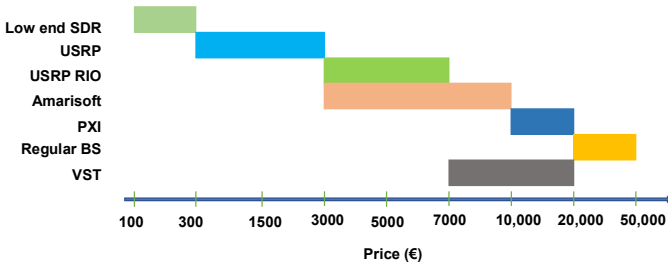


Fig. 1. Cost analysis of various RF transceivers

For example, $\mathcal{E}(f)$ of an RBS under clean channel conditions is around -110dBc/Hz @ 100kHz offset and for a low end SDR platform where the oscillator performance is usually compromised is around -70dBc/Hz @ 100kHz offset [12]. A detailed cost analysis as depicted in Fig. 1 shows that the

regular BS is a big financial investment with a very clean analog frontend characteristics whereas, the SDR platforms fall in different price categories comprising on analog frontend impairments including phase noise performance illustrated in Fig. 2.

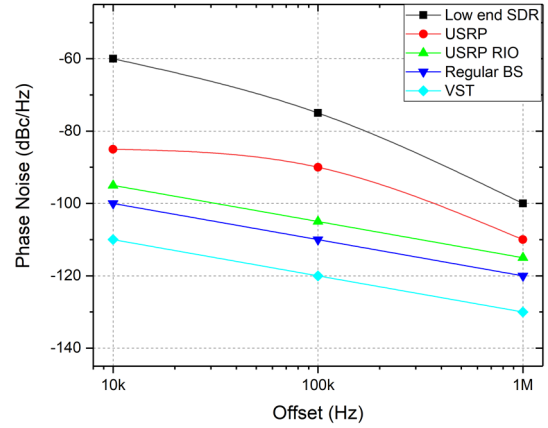


Fig. 2. $\mathcal{E}(f)$ analysis ranging from low end SDR to regular BS.

By analyzing this big $\mathcal{E}(f)$ performance difference between a potential SDR acting as an FBS and a legitimate BS, it is, therefore, proposed to implement an efficient FBS identification scheme on a mobile terminal that analyzes the amount of phase noise signatures. If $\tau_{\mathcal{E}}(t, f)$ is the threshold phase noise, then the criteria of detecting an FBS is defined by:

$$\mathcal{E}(f) \geq \tau_{\mathcal{E}}(t, f) \quad (2)$$

The $\tau_{\mathcal{E}}(t, f)$ is the function of time as well as frequency since the threshold value is proposed to be self-learned by the terminal itself through analyzing the $\mathcal{E}(f)$ of different RBS repeatedly over time in a normal mode of operation. From the analysis of Fig. 2, it is evident that low to medium price range of SDRs acting as FBS are easy to detect by defining an initial $\tau_{\mathcal{E}}(t, f)$ as -100dBc/Hz @ 100kHz . However, from the analysis, it is hard to define a detectable initial threshold $\tau_{\mathcal{E}}(t, f)$ for high performing SDR since there is no remarkable difference between an RBS and high end SDR acting as an FBS in terms of phase noise performance. Moreover, it is a well-established theory that in order to detect a phase noise, the phase noise performance of detecting terminal should be better than the transmitter's phase noise by at least 10-dB. In order to target the high end SDR, a more computationally efficient CFO based approach has been proposed in the next sub-section.

B. Carrier Frequency Offset Signatures

All central clocks in the legitimate cellular network of several RBSs are clock synchronized through precision time protocol (PTP) defined in IEEE 1588-2002. The NTP and GPS based synchronization suffers from serious interference and security concerns, so MNOs prefer a robust PTP based synchronization scheme that is based on master-slave hierarchy to lock on the grandmaster clock (GMC) e.g. atomic clock. The first-time stamp is generated by GMC which is based beyond the packet core network in the backhaul through routers and switches and RBSs being slaves derive their clocks by the GMC's first precise time stamp. In this manner, the whole

cellular network is highly clock synchronized both in frequency and time fashion [13 - 14].

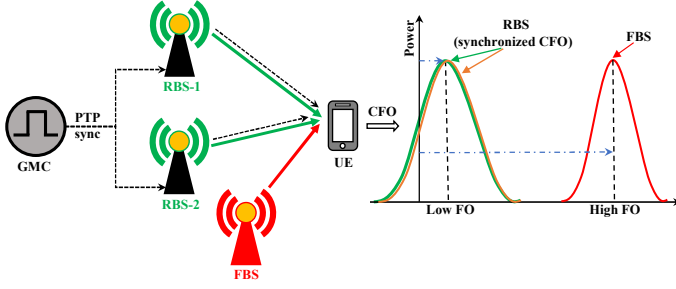


Fig. 3. Clock synchronous CFO based RF fingerprinting

In addition to the strong clock synchronization, an RBS clock is a costly and high precision device that gives low carrier frequency offset (CFO) values when measured at the mobile terminal. Whereas, the lousy clock generator in the FBS shows larger offset values at the detecting terminal.

Gaining benefit from the strong clock synchronization of cellular network and precise clock device of an RBS, it is speculated that CFO values of all the legitimate base stations when measured at a mobile terminal must nearly be identical. The FBS, on the other hand, is not hooked to the network synchronized clock gives larger and different offset values on the mobile terminal compared to RBSs, enabling a strong RF fingerprint to decide whether the serving BS is fake or legitimate. The CFO based identification scheme is shown in Fig. 3, where the mobile terminal estimates CFO of all surrounding base stations. If the terminal shows identical CFO for all the surrounding BS and different for just the serving cell, it means the serving cell is a fake cell powered by an FBS that is running freely by its own clock domain whether lousy or precise (GPS enabled).

CFO Estimation Techniques: The Fast Fourier Transform (FFT) approach is the most common one that computes the FFT of the incoming signal and calculates the offset by determining the signal peak. However, this method is computationally complex and requires substantial amount of processing time. In contrary to this, the coarse CFO method that operates on complex envelope $r(n)$ of the input waveform is computationally more powerful and time efficient [15 - 16]. Mathematically, the complex envelop is defined as:

$$r(n) = |r(n)|e^{j\theta_r(n)} \quad (3)$$

Then the coarse CFO estimate at sampling frequency f_s with phase offset $\Delta\theta_r$ is given by

$$\hat{\Delta f} = \frac{\Delta\theta_r}{2\pi} f_s \quad (4)$$

C. CFO Stability Signatures

The Stability is defined as the CFO as a function of time i.e. $\Delta f(t)$. The central clock of an RBS is highly precise and temperature-stabilized device, also called oven clock. Over the synchronization channels, the mobile terminal synchronizes to the BS. Due to bad quality of the clock in an FBS, it shows a

significant drift up and down of the carrier frequency offset value with time. The usage of stable and precise clock at the FBS stabilizes CFO, however the CFO magnitude is relatively large enough compared the synchronous CFO of the RBS. The concept is qualitatively presented in Fig. 4.

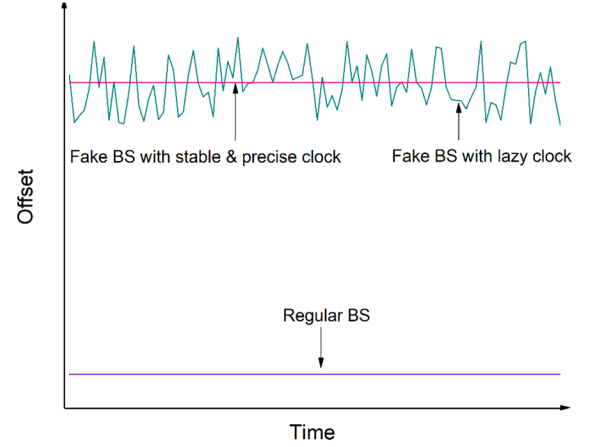


Fig. 4. CFO stability signatures

III. MEASUREMENT SETUP

A. Phase Noise Measurement Setup

The phase noise measurement setup is shown in Fig. 5, where different SDR platforms like USRP-B210, USRP-N210 (with SBX and XCVR frontend boards) and Adalm Pluto are mimicking as the FBS as high, medium and low ends respectively [17 - 18]. On the other side, the AnaPico APPH20G and FSUP R&S have been employed to act as mobile terminal to calculate the phase noise signatures [19 - 20].

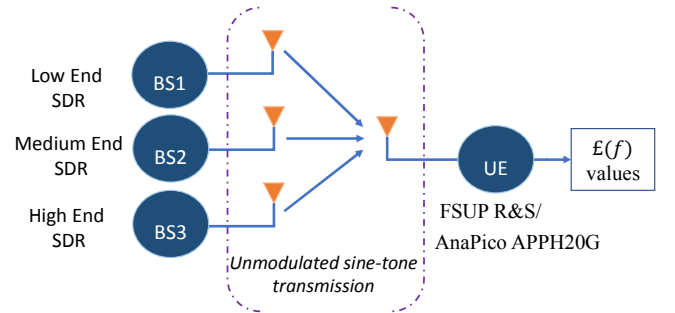


Fig. 5. Phase noise measurement setup

B. CFO Measurement Setup

The carrier frequency offset-based RF fingerprinting detection scheme has been verified in the lab environment by using two SDR platforms to mimic as the RBS where they are emulated using the PTP network-based clock synchronization as already shown in Fig. 3. For a highly stable clock precision, an external R&S clock source has been used to feed the two imitating SDR based RBS and to the terminal. Resultantly, a

complete lab based emulating RBS cellular network is established to gain from the concept of clock synchronization. The complete measurement setup is shown in Fig. 6 (a) & (b).

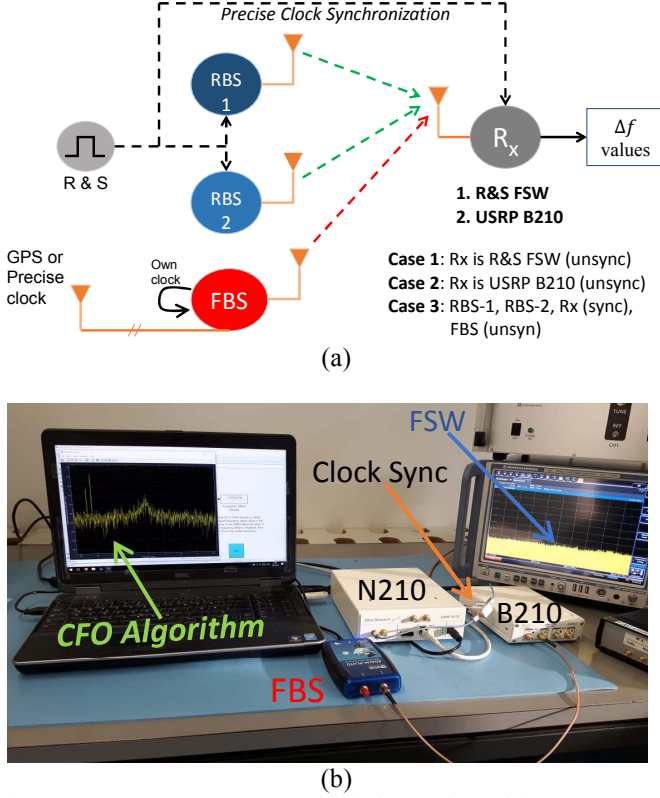


Fig. 6. (a) CFO measurement setup model, (b) actual CFO lab setup

IV. RESULTS & ANALYSIS

A. Phase Noise Results

Based on the measurement setup, phase noise results are shown in Fig. 7. The phase noise measurements show huge instability and oscillations in the low performing SDR platform that reveal the strong hardware impairments present in the frequency synthesizer part of the analog frontend. Based on the proposed threshold criteria i.e. $\tau_E(t, f)$, it is quite easy to detect such an FBS with highly instable internal oscillator. Whereas, the medium performing SDR platform which is N210 in the measurement setup shows a normal trend of phase noise, however, the $E(f)$ values at a defined initial $\tau_E(t, f)$ of -100dBc/Hz @ 100kHz , probability of detecting the medium end FBS is still very high. Finally, looking at the trend of high performing SDR i.e. B210, the probability of detection using the phase noise signatures is quite low since the trend just rovers around the defined threshold.

B. CFO Results

To overcome the issues offered by phase noise approach, the setup defined in Fig. 6 to attain the CFO based RF fingerprints is very robust and efficient to detect fake cells. In contrary to phase noise approach, where the terminal's own phase noise must be better than the transmitter's phase noise, it has been

proved in the next sections that validating the CFO based detection doesn't require a high-quality receiver. In order to prove this concept supported by results, following three cases have been developed to deduce some important conclusions.

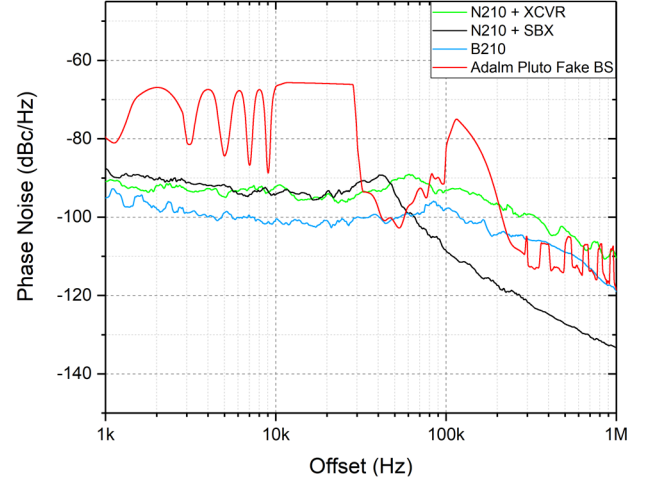


Fig. 7. Phase noise measurement results

Case-I (Rx is R&S FSW): At first, a high-quality R&S FSW has been used as the receiver terminal. In this case, all the SDR nodes have intentionally been set to be running independently by their own clock domains. The CFO as a function of whole sub-6GHz range of carriers have been generated in Fig. 8 (a). The trend for the low-end SDR shows huge offset values as the carrier frequency increases.

Case-II (Rx is USRP B210): By changing the receiver from a measurement equipment to an SDR platform B210 mimicking the real mobile terminal, shows a similar trend in offset values of other SDR platforms i.e. USRP N210 and Adalm Pluto as revealed in Fig. 8 (b). This confirms that the mobile terminal doesn't necessarily have to be the high performing and computationally powerful as was the case with phase noise signatures.

Case-III (Synchronized RBS-1, 2 and Rx): In this case, measurement model presented in Fig. 6 (a) has been implemented by imitating N210 and B210 SDR platforms as RBS and feeding them with a high precision stable clock from an R&S clock source that achieves the synchronization among the emulating network within a lab environment. Whereas, Adalm Pluto running by an independent clock domain attempts to attack the terminal. The terminal attached to the laptop computes the CFO values. The results at 800-MHz carrier frequency illuminated in Fig. 8(c) clearly depicts that the synchronized RBSs show identical CFO values at the receiver which is found to be 2.344 Hz, whereas, the attacker running on another clock domain and based on strong hardware impairments show large offset values. This difference can surely be utilized by a terminal to detect the FBS. The former Case-I & II confirms that as long as all the SDR platforms (including medium and high end) are running independently by their individual clock domains, they show large offset values from several hundreds of Hz to KHz. As soon as the same SDR nodes are turned to synchronized and precise clock domain, the CFO

comes out to be identical for all the nodes. Hence, CFO based RF fingerprinting have proven to be quite promising and efficient technique for detecting FBS.

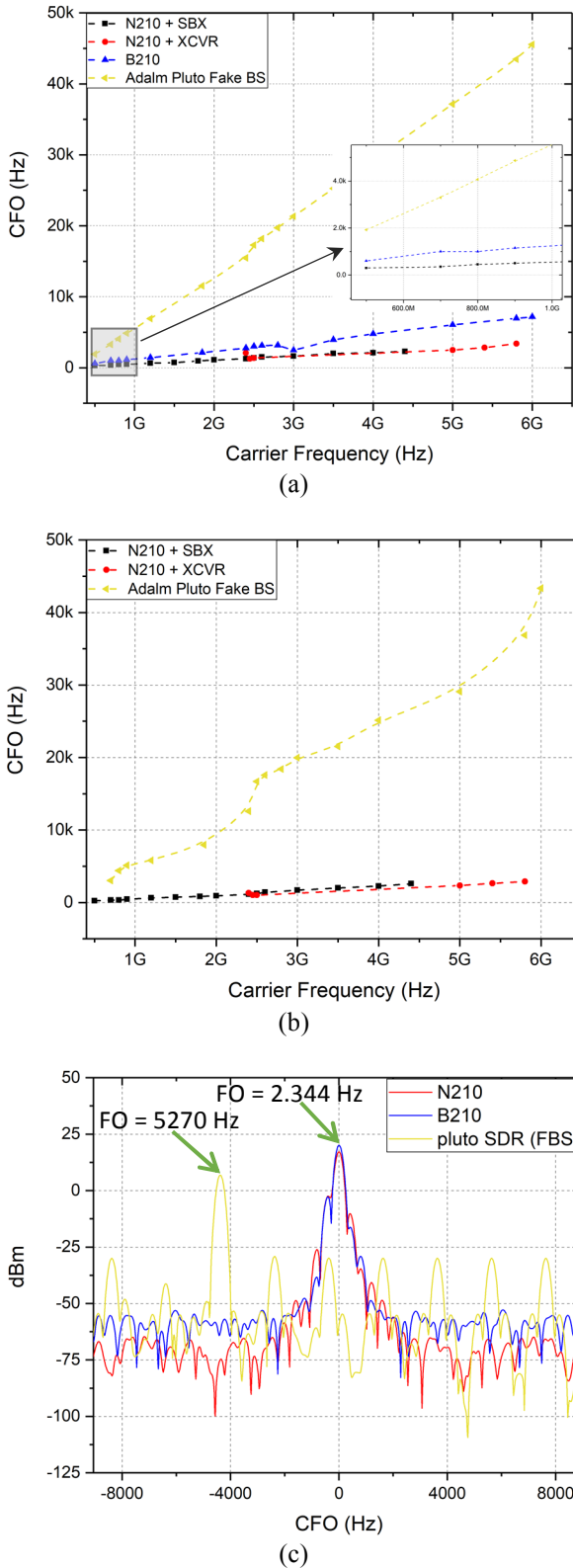


Fig. 8. (a) Case-I: when Rx is R&S FSW, (b) Case-II: when Rx is USRP B210, (c) Case-III: when RBS-1, 2 and Rx are synchronized under FBS attack.

C. CFO Stability Results

CFO stability criteria has proven to be even more powerful than the CFO based detection scheme described with results in the previous sub-section. In this sub-section, it has been revealed that when medium to high end SDR platforms are configured as FBS then the combination of CFO and CFO stability-based detection guarantees catching an FBS. Fig. 9 (a) illustrates this idea. At first, when N210 is configured to act as an FBS with an internal lousy clock domain, the CFO stability shows random behavior, whereas when it is fed by a precise clock, yet it still belongs to different clock domain as terminal and hence it shows remarkable improvement in CFO stability. Nevertheless, this stable CFO is still large enough to decide exclusively that serving BS is a fake one. Interestingly, when the similar SDR node N210 is brought to the same network clock domain as the terminal do, then its CFO detected at the terminal drops to 2.334 Hz. The same behavior has been observed for the high performing SDR node B210 and all other tested SDR platforms that if they are operated at their individual clock domains, they show either instable or stabilized larger offsets and as soon as they are brought to the network clock domain, their CFO values fall down massively as shown in Fig. 9 (b).

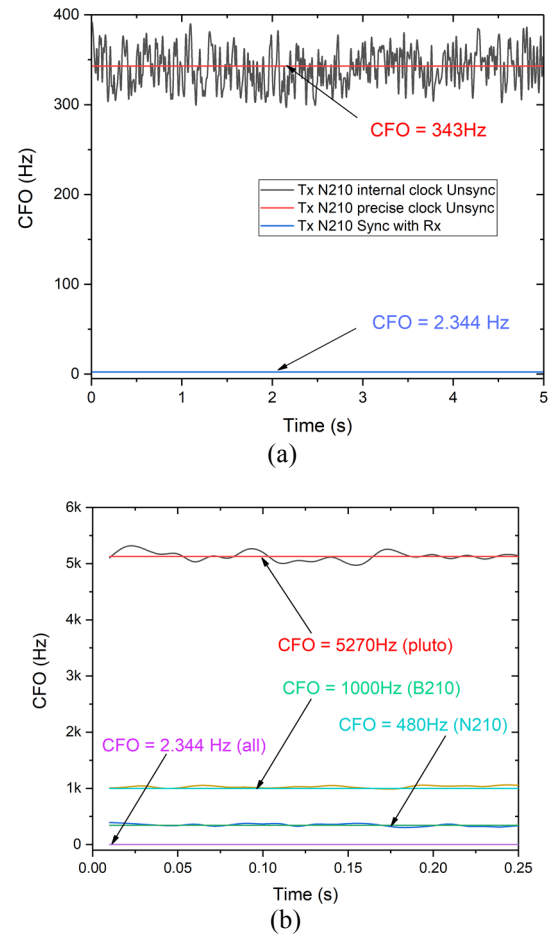


Fig. 9. CFO stability when (a) N210 is FBS (b) all SDRs act as FBS at individual clock domain

V. CONCLUSION

The paper has presented very efficient RF fingerprinting techniques to detect the fake base station (FBS) in a cellular network. The first approach deals with measuring the phase noise signatures of received signal at the mobile terminal. If the measured phase noise is greater than the defined initial threshold $\tau_E(t, f)$, the terminal detects fake cell. With the aid of measurement results, it has been established that low to medium performing SDR platforms acting as FBS are extremely probable to be detected, while the probability of detecting the high end SDRs is quite low with this approach. To tackle this problem, concept of cellular network-based synchronization has been profited to propose a highly efficient carrier frequency offset based FBS detection scheme. With the help of lab-based measurement setup, it has been shown that the regular base stations (RBS) that reside on the same clock domain as the terminal, show synchronous and nearly identical CFO values and the FBS that is fed by its independent clock shows large and instable CFO trend. Furthermore, with the help of CFO stability criteria, it has been shown that time dependent instabilities even tend to detect the high end SDR platforms acting as FBS, if clock synchronization of RBS network is present.

The current and future extension of this work involves determining further unique RF characteristics of RBS such as Power Amplifier (PA) non-linearities and the digital predistortion. The implementation of digital pre-distortion to tackle PA non-linearities is quite expensive and not spent in any present SDR platforms to the best of our knowledge. PA pre-distortion-based RF fingerprints will be explored in the follow-up publication in terms of statistical higher order moments (HOM) to implement the robust detection schemes in terms of reliability. Additionally, an effort shall be made in determining how noise of the RBS is shaped compared to FBS followed by capturing of RF signatures from an actual base station in field trials. Finally, an error vector based statistical fingerprinting will be proposed to detect FBS on an actual mobile terminal.

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